Titanic Survival Prediction: A Binary Classification Study

Comparing Logistic Regression, Random Forest, and SVM on the Titanic passenger dataset

1. Introduction

This report summarizes a binary classification study predicting whether a Titanic passenger survived the 1912 sinking, using demographic and ticketing attributes (class, sex, age, fare, family size, port of embarkation). The Titanic dataset (891 passengers, sourced from the public Kaggle/seaborn-data mirror) is a well-documented benchmark with realistic data-quality issues — missing values, a mix of numeric and categorical fields, and a moderately imbalanced target (62% died, 38% survived) — making it well suited to demonstrate a complete preprocessing-to-deployment classification workflow

2. Data Preprocessing & EDA

Six columns were dropped for leaking the target or duplicating other fields (alive, class, who, adult_male, embark_town) or being too sparse to impute reliably (deck, 77% missing). Missing age values (177 rows) were imputed using the median age within each (passenger class, sex) group, which is more informative than a single global median since age varies meaningfully across those groups; the two missing embarked values were filled with the mode. sex and alone were binary-encoded and embarked was one-hot encoded. The data was split 80/20 into train/test sets, stratified to preserve the survival rate in both

EDA confirmed the historical narrative of the disaster: sex (r ≈ 0.54) and fare/class are the strongest correlates of survival, survival skews younger, and 1st-class passengers survived at far higher rates than 3rd-class passengers (Figure 1). Because the target is imbalanced, accuracy alone is not a sufficient evaluation metric — a model predicting "died" for everyone would score ~62% accuracy while being useless — so precision, recall, F1, and ROC-AUC were all tracked

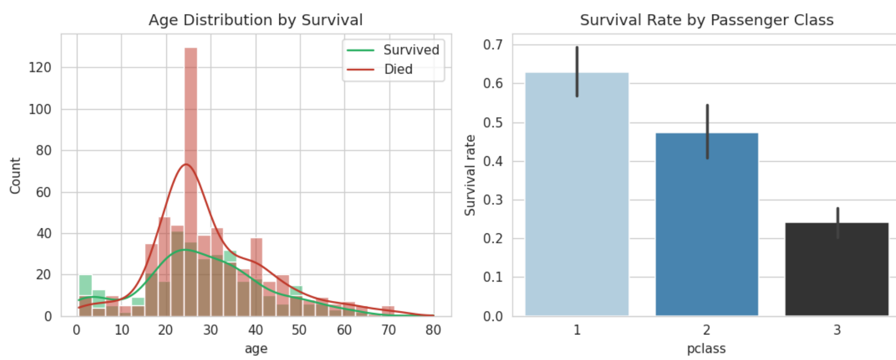

Figure 1. Survival skews younger; survival rate drops sharply as cabin class declines.

3. Model Training, Evaluation & Interpretation

Three classifiers were trained: Logistic Regression (linear, interpretable baseline), Random Forest (non-linear, handles feature interactions), and an SVM with an RBF kernel (margin-based, non-linear). Logistic Regression and SVM inputs were standardized; Random Forest used the raw encoded features, as tree ensembles are scale-invariant

Model	Accuracy	Precision	Recall	F1	ROC-AUC
SVM (RBF)	0.810	0.818	0.652	0.726	0.849
Logistic Regression	0.804	0.783	0.681	0.729	0.849
Random Forest	0.804	0.774	0.696	0.733	0.835

All three models perform comparably (80–81% accuracy, 0.835–0.849 ROC-AUC), but no single model dominates on every metric: SVM has the best precision and ties Logistic Regression on ROC-AUC but has the lowest recall — it misses more true survivors than the other two. Random Forest has the best recall and F1, the most "balanced" choice on this moderately imbalanced target, at a small cost to precision. The right choice therefore depends on deployment priorities: minimizing missed survivors (favoring Random Forest's recall) versus minimizing false "survived" calls (favoring SVM's precision)

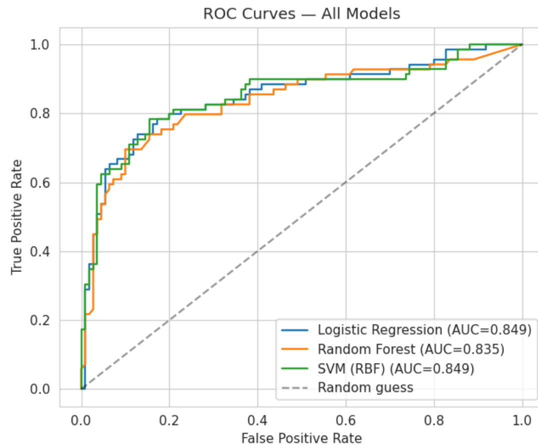

Figure 2. ROC curves — SVM and Logistic Regression edge out Random Forest on ranking quality (AUC), though all three are close.

Random Forest feature importances and Logistic Regression coefficients agree on the dominant predictors: sex, fare, and passenger class outweigh family-size variables (sibsp, parch, alone) and embarkation port, consistent with the "women and children first" / class-based lifeboat access narrative of the disaster

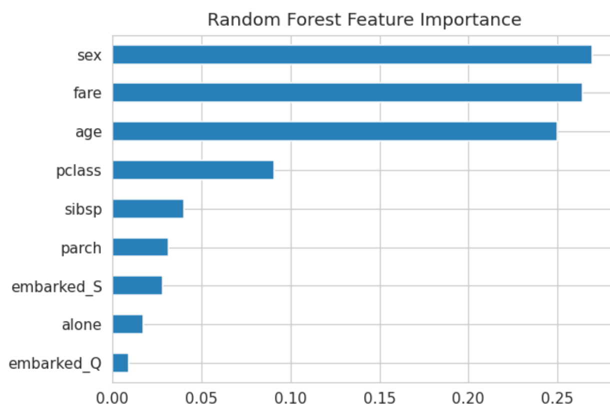

Figure 3. Random Forest feature importances — sex, fare, and age dominate; family-size and embarkation features contribute comparatively little.

4. Deployment & Monitoring

Deployment: package the chosen model together with its exact fitted preprocessing (imputation rules, scaler, encoder) as a single versioned artifact (e.g., a serialized scikit-learn Pipeline), serve it behind a containerized REST API (FastAPI/Flask + Docker) with a /predict endpoint, and deploy to a managed, independently scalable endpoint

- Key production risks: schema drift (missing fields at request time), unseen categories breaking one-hot encoding, train/serve skew if preprocessing logic is duplicated rather than shared, and inference latency for ensemble/SVM models under high load
- Monitoring strategy: track input feature distributions for drift (e.g., PSI/KS-tests), monitor the distribution of predicted probabilities for sudden shifts, periodically recompute accuracy/F1/AUC once ground truth becomes available, log every prediction with its model version for auditability, and retrain on a fixed cadence or upon detected drift using a champion/challenger evaluation before promotion

5. Conclusion

All three models are viable, with accuracy in the low-to-mid 80% range and ROC-AUC of 0.84–0.85. Sex, passenger class, and fare are consistently the strongest predictors across both the linear and non-linear models. Model selection should be driven by the real-world cost of false positives versus false negatives in the target deployment context, rather than by a single aggregate metric